

HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 16: System Validation

Version: 1.00

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1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	15:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit.
ADC	Analog to digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.
QM	Quality Management level below ASIL.
SIL	Safety Integrity Level.
SW	Software.

4 System Validation

The main objective is to prove that the Airbag ECU does not activate the airbags unintentionally, deploys the required airbag output only when a valid crash and safing condition are detected, suppresses the passenger airbag when the baby-seat input is active, and enters a defined safe state when a fault is detected.

4.1 Safety Goals and Safety Parameters

Tabelle 2 Safety Goals and Safety Parameters

ID	Safety Goal	ASIL	Safe State	Main Safety Parameters
SG1	Prevent unintended driver/passenger airbag deployment.	ASIL D	Deployment inhibited; warning lamp ON if fault exists.	Fail-safe behavior, diagnostic coverage, fault detection, output readback, independence.
SG2	Deploy required airbag(s) during a valid crash within the allowed reaction time.	ASIL D	If deployment cannot be guaranteed, fault shall be indicated and stored.	Availability, deterministic timing, FTTI/PST, redundancy, backup energy.
SG3	Suppress passenger airbag when baby seat/passenger suppression condition is active.	ASIL C	Passenger airbag OFF; driver airbag allowed if valid crash.	Safe input, input plausibility, priority logic, fail-safe suppression.
SG4	Activate the correct airbag output according to impact type and valid sensor combination.	ASIL D	Wrong output inhibited; fault indicated if selection is inconsistent.	Sensor plausibility, redundancy, cross-check, deterministic logic.
SG5	Detect and report sensor, output, CAN, memory, timing, or watchdog faults.	ASIL D support	Airbags OFF unless valid crash and safing path are confirmed; warning lamp ON.	DC, SFF support, fault containment, traceability, testability.

4.2 Test Report Template

Tabelle 3 Test Report Template

Field	Entry
Test ID	
Date / Tester	
Requirement(s)	
Safety parameter verified	
ASIL/SIL relevance	
Initial conditions	
Input values / sequence	
Expected result	
Actual result	
Evidence file name	
Pass / Fail	
Deviation / corrective action	
Retest result	

4.3 Example Test Execution Log

Tabelle 4 Example Test Execution Log

Test ID	Expected	Actual	Result	Evidence
SV-01	No deployment, lamp OFF.	To be filled during execution.	Open	serial_log_SV01.txt
CR-01	Driver ON, Passenger ON, CAN crash status.	To be filled during execution.	Open	vid-eo_CR01.mp4 / can_CR01.log
BS-01	Driver ON, Passenger OFF.	To be filled during execution.	Open	video_BS01.mp4
FI-01	Airbags OFF, warning lamp ON, CAN fault.	To be filled during execution.	Open	serial_FI01.txt

4.4 Final Validation Statement

The Airbag ECU prototype is expected to prevent unintended deployment, to deploy airbag outputs only when a valid crash and safing condition exists, to suppress the passenger airbag while the baby-seat is active, to enter a fail safe state when a fault is detected and to report its status through a CAN communication bus per the original design's specifications.

Final validation cannot be accomplished until the planned test cases have been executed and objective evidence such as logs, screenshots, CAN trace files and measurement records have been collected. All remaining limitations and open issues may be found in Phase 15, Section 4.6.

4.5 Review and Approval

Tabelle 5 Review and Approval

Role	Name	Date	Signature / Comment
Project Lead			
Functional Safety Manager			
Software Developer			
Hardware Developer			
Test Engineer			
Quality/Documentation Responsible			

5 Figure and table listing

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6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Airbag ECU Testing and Validation Plan	V3	31.05.2026	LCP13_Airbag_ECU_Testing_and_Validation_Plan_V3.pdf
2	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf
3	Arduino Coding and Logic Implementation for Airbag ECU System	V1	02.06.2026	FuSa Report_Adruino.docx
4	Airbag ECU ASIL Code Division	V02	02.06.2026	Airbag_ECU_ASIL_Code_Division_V02.docx
5	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

- [1] International Organization for Standardization, ISO 26262-3:2018, Road vehicles - Functional safety - Part 3: Concept phase. Geneva: ISO, 2018. <https://www.iso.org/standard/68385.html>
- [2] International Organization for Standardization, ISO 26262-4:2018, Road vehicles - Functional safety - Part 4: Product development at the system level. Geneva: ISO, 2018. <https://www.iso.org/standard/68386.html>
- [3] International Organization for Standardization, ISO 26262-6:2018, Road vehicles - Functional safety - Part 6: Product development at the software level. Geneva: ISO, 2018. <https://www.iso.org/standard/68388.html>
- [4] International Organization for Standardization, ISO 26262-8:2018, Road vehicles - Functional safety - Part 8: Supporting processes. Geneva: ISO, 2018. <https://www.iso.org/standard/68390.html>
- [5] International Electrotechnical Commission, IEC 61508-1:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 1. Geneva: IEC, 2010.
- [6] International Electrotechnical Commission, IEC 61508-3:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 3: Software requirements. Geneva: IEC, 2010.
- [7] ISO/IEC/IEEE, ISO/IEC/IEEE 29119-1:2022, Software and systems engineering - Software testing - Part 1: General concepts. Geneva: ISO, 2022.
- [8] International Organization for Standardization, ISO 11898-1:2024, Road vehicles - Controller area network (CAN) - Part 1: Data link layer and physical coding sublayer. Geneva: ISO, 2024.
- [9] AUTOSAR, Specification of RAM Test, AUTOSAR Classic Platform R24-11. AUTOSAR, 2024. <https://www.autosar.org>
- [10] MISRA Consortium, MISRA C:2025 - Guidelines for the use of the C language in critical systems. MISRA, 2025. <https://misra.org.uk>

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_16	14.06.2026	v1.00 Initial version. System Validation compiled from Testing Plan V3 Sections 4, 13, 14, 16, 17.